



Graduation Project Proposal

Project Information		
Project Title	Department/Faculty/University	Project Field/Discipline
Game Forge: An MCP-Based Automated Game Generation Server	Department of Computer Science/Faculty of Computers and Artificial Intelligence/Cairo University	Computer Science/Software Engineering and Artificial Intelligence
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If the project is sponsored by or initiated by an ICT company, please state its name:



Motivation

Please write why you chose this project idea, explaining clearly the problem that the project is addressing

Game development is one of the most complex domains in software engineering, requiring the integration of programming, engine expertise, art, music, design, and continuous creative iteration. This complexity makes it particularly challenging for solo developers and small teams to coordinate all required skills and produce market-ready products. As a result, many promising game ideas fail to progress beyond the concept stage or a minimal prototype, limiting their ability to attract investors, collaborators, or community support.

Recent advances in large language models and generative technologies have made it possible to automate significant portions of the game creation pipeline, including high-level design, code generation, and asset creation. However, existing tools often address these components in isolation and lack an integrated workflow that produces coherent, playable prototypes.

*We chose this project because it allows us to leverage our background in computer science, software engineering, artificial intelligence, and game development to build **Game Forge**, a practical system that lowers the barrier to entry for game development by transforming natural-language prompts into playable game prototypes.*



Project Description

Provide a brief description of the project in no more than 200 words.

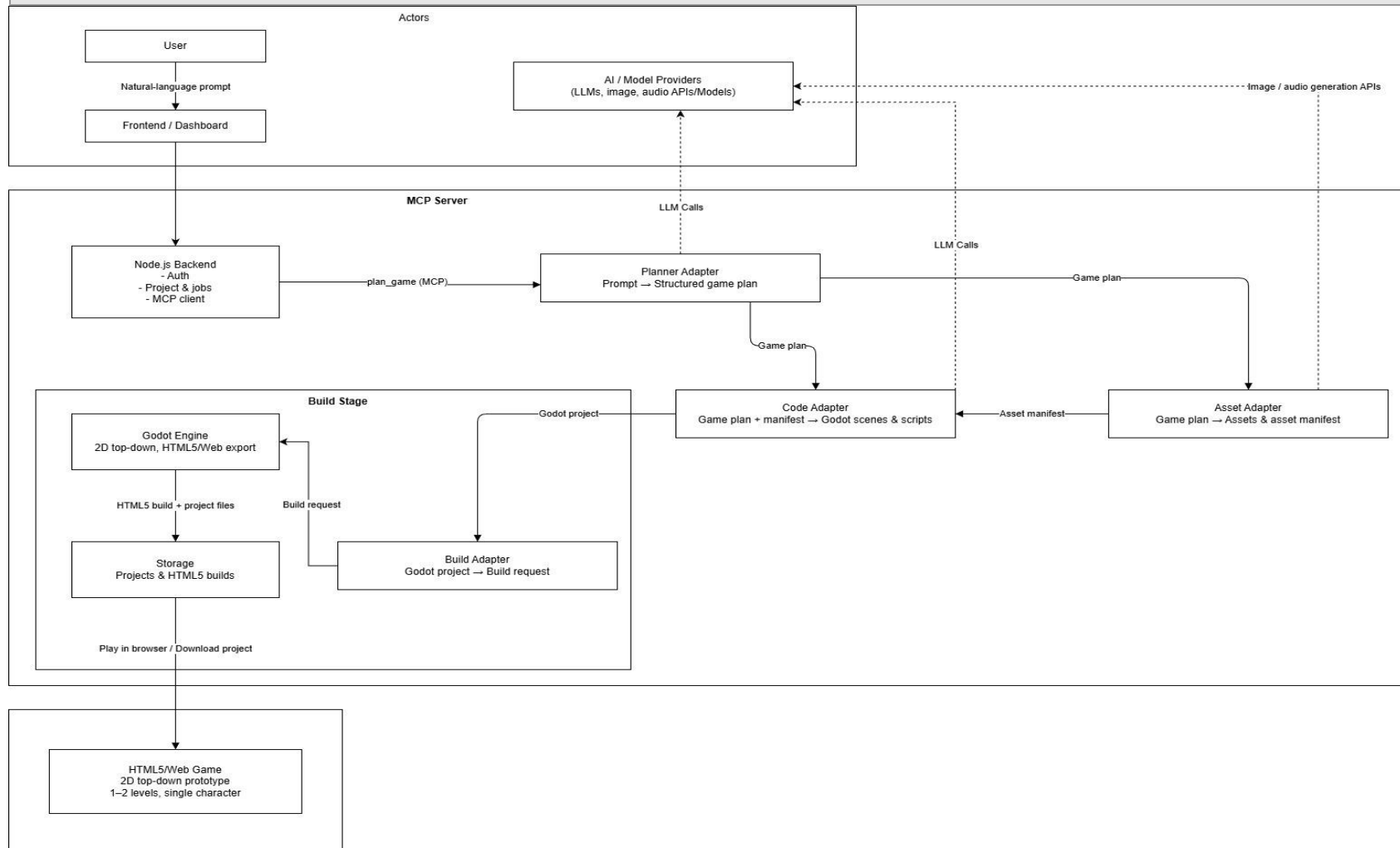
Game Forge is an MCP-based (Model Context Protocol) automated game generation system that transforms natural language prompts into content game prototypes. It targets HTML5/Web builds generated with the Godot engine, enabling users to play the resulting games in the browser and download the Godot project for further customization.

The system architecture is centered on a Node.js backend responsible for authentication, management, and job orchestration. This backend acts as an MCP client that coordinates the workflow by communicating with a single MCP server. The server exposes four adapters: a Planner adapter that converts prompts into structured game designs; an Asset adapter that prepares or generates basic visual and audio assets; a Code adapter that produces Godot scenes and scripts; and a Build adapter that invokes the Godot engine to export an HTML5 build and store both the project and the compiled game.

The initial version of Game Forge focuses on generating simple 2D top-down games with one to two levels, a single playable character, basic interactions, and simple win/lose conditions. These games are generated from user prompts and exported as playable HTML5 builds alongside editable Godot project files, with future extensions planned for additional genres, richer assets, and alternative input modalities.

Block Diagram

Please insert the expected or envisioned project detailed block diagram below. This diagram should reflect the main workflow of the system, including its inputs, how they progress through the system, and the expected outputs.





Project Plan

Please define the approach and phases to deliver the intended project outcome.



Game Forge adopts Agile with 2-week sprints for iterative development, enabling rapid prototyping and feedback on core features.

Phase 0: Design and Setup

Goals: Determine project scope and prepare the development environment.

Tasks: Finalize system architecture, create system diagrams, set up the GitHub repository, define the folder structure, configure environment variables, and install required dependencies.

Deliverables: project architecture, complete folder structure, initialized GitHub repository, and system design documentation.

Phase 1: Core Platform Development

Goals: Build the foundational web platform and data layer.

Tasks: Implement backend server using Node.js and Express, connect MongoDB, define user and project schemas, and create basic frontend pages for prompt submission and project management.

Deliverables: Web application with prompt input, project creation, project storage, and a basic user dashboard.

Phase 2: Game Planning (AI Reasoning)

Goals: Transform natural language prompts into structured game design specifications.

Tasks: Integrate a planner language model, define a strict JSON schema for game plans, validate generated outputs, and store structured plans in the database.

Deliverables: Planner service that converts user prompts into validated game design JSON objects.



Phase 3: MCP Command Generation

Goals: Convert game design specifications into deterministic execution commands.

Tasks: Design the MCP command schema, implement command generation logic, map game elements to MCP commands, and validate commands before execution.

Deliverables: A validated list of MCP commands representing the full game creation process.

Phase 4: MCP Server Implementation

Goals: Safely execute MCP commands and translate them into engine-level actions.

Tasks: Implement a single MCP server, define command handlers, create internal adapters for assets, code, and builds, and log all command executions.

Deliverables: A functional MCP server capable of executing validated commands.

Phase 5: Asset Generation

Goals: Automatically generate visual and audio assets for the game.

Tasks: Integrate image and audio generation models, generate sprites, backgrounds, and sound effects, and store generated assets.

Deliverables: Generated and stored game assets associated with each project.

Phase 6: Godot Project Assembly and Build

Goals: Construct and export a playable game using the Godot engine.



Tasks: Programmatically create Godot scenes and nodes, attach generated scripts and assets, and export the project to HTML5/Web Assembly.

Deliverables: A playable web-based game build

Phase 7: System Integration and Execution Pipeline

Goals: Integrate all system components into a single automated workflow.

Tasks: Connect frontend, backend, language models, MCP server, and Godot pipeline, and implement job tracking and execution status updates.

Deliverables: An end-to-end pipeline that generates a playable game from a single user prompt.

Phase 8: Testing, Optimization, and Deployment

Goals: Ensure system reliability and performance.

Tasks: Perform functional testing, handle edge cases, optimize execution flow, improve error handling, and deploy the system.

Deliverables: Stable deployed system ready for demonstration.

Phase 9: Documentation and Defense Preparation

Goals: Prepare academic documentation and presentation materials.

Tasks: Write the final project report, document system architecture, and MCP workflow, prepare diagrams, and create a demonstration walkthrough.

Deliverables: Final report, presentation slides, system diagrams, and live demo.



Prototype Prospects

List the Egyptian ICT companies or beneficiaries that may be interested in the developed prototype and the end-users/customers (name the specific class of individuals, governmental agencies, ministries ... etc. that will benefit from the prototype)

List of ICT Companies and/or Beneficiaries:

- Rumbling Games (Cairo)—AAA studio; uses rapid Arabic game prototyping.
- Mirrorful (Cairo)—Game specialists; integrates digital experience tools.
- Mobisoft Solutions (Cairo)—Game dev services; adopt client project efficiency.
- ECO TECH EFAD (Cairo)—IT/game consultants; employs custom tool development.
- GeekyAnts Egypt (Cairo)—Software/game innovators; leverages for UI/UX prototyping.
- ITIDA (Gov agency)—Funds ICT prototypes via hackathons like Digitopia.

Potential End-Users/Consumers:

- MCIT (Ministry of Communications & IT)—Hackathons, gaming innovation pilots.
- Ministry of Higher Education – University ICT labs, student game projects.
- Ministry of Education—Edutainment apps for digital curricula.
- Egypt University of Informatics (EUI)—Game dev students/faculty prototyping.
- TIEC (Tech Innovation Center)—Startup accelerators for game tools.
- Indie developers—low-cost asset generation.